

IOWA STATE UNIVERSITY

LAB 9 - PRESSURE BULKHEAD REPORT

LUCAS TAVARES VASCONCELLOS

PROFESSOR

DR. VINAY DAYAL

College of Engineering

Aerospace Engineering

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CONTENTS

Contents	i
1 Introduction	1
2 Figures and Tables	2
3 Methodology and Results	3
4 Conclusion	6
A Appendix A	7
A.1 Rough Work	7

INTRODUCTION

In this lab, we designed an ellipsoidal pressure bulkhead for a fuselage to withstand high-altitude pressurization. The shell, constructed from a quasi-isotropic Graphite/Epoxy composite, was designed to withstand these stresses with a factor of safety of at least 4. We calculated the tensile, compressive, and shear stresses at both the top and equator of the bulkhead to verify that the composite material meets the maximum allowable stress limits under these conditions. Additionally, we also analyzed two connection options: riveting and adhesive bonding.

FIGURES AND TABLES

Fastener Diameter		3/32	1/8	5/32	3/16	1/4	5/16	3/8	7/16	1/2	9/16
Fastener Material	F_{su} (Ksi)	Ultimate Single Shear Load (lb/fastener)									
1100F	9										
5056(B)	28	203	363	556	802	1450	2290	3280			
2117-T3(AD)	30	217	388	596	862	1550	2460	3510			
2017-T31(D)	34	247	442	675	977	1760	2790	3970			
2017-T3(D)	38	275	494	755	1090	1970	3110	4450			
2024-T31(DD)	41	296	531	815	1180	2120	3360	4800			
Monel	49	355	635	973	1400	2540	4020	5730			
7075-H75	38	275	494	755	1090	1970	3110	4450			
A-286 CRES	90	—	—	1726	2485	4415	6906	9938			
Ti-6Al-4V & Alloy Steel	95	—	—	1822	2623	4660	7290	10490	14280	18650	23610
Alloy Steel	108	—	—	2071	2982	5300	8280	11930	16240	21210	26840
Alloy Steel	125	—	—	2397	3452	6140	9590	13810	18790	24540	31060
H-11 Steel	132	—	—	2531	3645	6480	10120	14580	19840	25920	32800

(Refer to MIL-HDBK-5D per Ref. 7.6)

Fig. 7.1.3 Fastener shear-off allowable loads.

Table 2.1: Fastener shear-off allowable loads

METHODOLOGY AND RESULTS

For this week's lab, we were assigned to design an ellipsoidal shell for the pressure bulkhead of an aircraft. We are to consider the design ceiling for a Boeing 787 fuselage of 13100 m. The first step was to calculate P_{in} , P_{out} , and the difference in pressure ΔP :

$$P_{\text{in}} = 0.8 \cdot (101.325 \text{ kPa}) = 81.06 \text{ kPa} \quad (3.1)$$

$$T = 15.04 - 0.00649 \cdot (13100 \text{ m}) = -69.979 \quad (3.2)$$

$$P_{\text{out}} = (101.325 \text{ kPa}) \left(\frac{-69.979 + 273.1}{288.08} \right)^{5.25588} = 16.147 \text{ kPa} \quad (3.3)$$

$$\Delta P = P_{\text{in}} - P_{\text{out}} = 81.06 \text{ kPa} - 16.147 \text{ kPa} = 64.913 \text{ kPa} \quad (3.4)$$

The problem states that the shell is made of a quasi-isotropic Graphite/Epoxy composite with $E = 62 \text{ GPa}$ and bending stiffness $E_b = 200 \text{ GPa}$. Each ply has a thickness of $t = 125 \times 10^{-6} \text{ m}$. Thus, our design has a configuration $[0^\circ, -45^\circ, +45^\circ, +90^\circ]_s$, thus our total thickness is $t = 0.001 \text{ m}$.

We also know from the problem statement the maximum stresses in tension, compression, and shear: $\sigma_T = 290 \text{ MPa}$, $\sigma_C = 240 \text{ MPa}$, and $\tau_{\text{max}} = 160 \text{ MPa}$. Additionally, we need a factor of safety of at least 4 for all components. a (base radius) is provided as 1.8 m. b (vertical component of ellipse) was calculated from the relation of maximum stress in tension with the stress in tension at the top of the pressure bulkhead. Then, the stresses and factors of safety were calculated at the top and at the Equator.

Top of the shell:

$$b = \text{FS} \cdot \frac{\Delta P \cdot a^2}{2 \cdot \sigma_T \cdot t} = 4 \cdot \frac{(64.913 \text{ kPa})(1.8 \text{ m})^2}{2(290 \times 10^3 \text{ kPa})(0.001 \text{ m})} = 1.450 \text{ m} \quad (3.5)$$

$$\sigma_\phi = \sigma_\theta = \frac{(64.913 \text{ kPa})(1.8 \text{ m})^2}{2(1.450 \text{ m})(0.001 \text{ m})} = 72.500 \text{ MPa} \quad (3.6)$$

$$\sigma_3 = -P_{in} = -0.0811 \text{ MPa} \quad (3.7)$$

$$\tau = \frac{(72.500 \text{ MPa}) - (-0.0811 \text{ MPa})}{2} = 36.291 \text{ MPa} \quad (3.8)$$

$$FS_{\sigma_\phi} = FS_{\sigma_\theta} = \frac{290 \text{ MPa}}{72.500 \text{ MPa}} = 4.00 \quad (3.9)$$

$$FS_{\sigma_3} = \frac{240 \text{ MPa}}{0.0811 \text{ MPa}} = 2960 \quad (3.10)$$

$$FS_\tau = \frac{160 \text{ MPa}}{36.291 \text{ MPa}} = 4.41 \quad (3.11)$$

Equator of the shell:

$$\sigma_\phi = \frac{(64.913 \text{ kPa})(1.8 \text{ m})}{2(0.001 \text{ m})} = 58.422 \text{ MPa} \quad (3.12)$$

$$\sigma_\theta = \frac{(64.913 \text{ kPa})(1.8 \text{ m}) \left(1 - \left(\frac{(1.8 \text{ m})^2}{2(1.450 \text{ m})^2} \right) \right)}{(0.001 \text{ m})} = 26.814 \text{ MPa} \quad (3.13)$$

$$\sigma_3 = -P_{in} = -0.0811 \text{ MPa} \quad (3.14)$$

$$\tau = \frac{(58.422 \text{ MPa}) - (-0.0811 \text{ MPa})}{2} = 29.252 \text{ MPa} \quad (3.15)$$

$$FS_{\sigma_\phi} = \frac{290 \text{ MPa}}{48.422 \text{ MPa}} = 4.96 \quad (3.16)$$

$$FS_{\sigma_\theta} = \frac{290 \text{ MPa}}{26.814 \text{ MPa}} = 10.8 \quad (3.17)$$

$$FS_{\sigma_3} = \frac{240 \text{ MPa}}{0.0811 \text{ MPa}} = 2960 \quad (3.18)$$

$$FS_\tau = \frac{160 \text{ MPa}}{29.525 \text{ MPa}} = 5.47 \quad (3.19)$$

For the option of using rivet to connect the shell, we start the design calculating the number of rivets required to keep a 5 cm spacing:

$$N^\circ \text{ of Rivets} = \frac{2\pi(1.8 \text{ m})}{(0.05 \text{ m})} = 226.2 \Rightarrow 227 \text{ rivets} \quad (3.20)$$

With the quantity of rivets defined, we calculated the single shear load $F_{\text{rivet},\tau}$ and applied the required factor of safety of 4 to choose a rivet design.

$$F_{\text{rivet},\tau} = \frac{\pi(1.8 \text{ m})^2(64.913 \text{ kPa})}{(227 \text{ rivets})} = 2.911 \text{ kN/rivet} = 654.419 \text{ lbs/rivet} \quad (3.21)$$

$$F_{\text{rivet},\tau,\text{FS}} = 4 \cdot 654.419 \text{ lbs/rivet} = 2617.68 \text{ lbs/rivet} \quad (3.22)$$

From [Table 2.1](#), we chose the fastener with $D = 3/16$ in, Ti-6Al-4V & Alloy Steel material, with $F_u = 2623$ lbs.

Thus, the factor of safety is:

$$\text{FS}_\tau = \frac{F_u}{F_{\text{rivet},\tau}} = \frac{2623 \text{ lbs}}{654.19 \text{ lbs}} = 4.01 \quad (3.23)$$

Lastly, we were also asked to consider connecting the bulkhead using a 3M Scotch-Weld Structural Adhesive DP-8005. It's shear strength is of 17.05 MPa. Thus we calculated shear at the adhesive and its factor of safety:

$$\tau_{\text{adhesive}} = \frac{\pi \cdot a^2 \cdot \Delta P}{2 \cdot \pi \cdot a \cdot w_{\text{adhesive}}} = \frac{a \cdot \Delta P}{2 \cdot w_{\text{adhesive}}} = \frac{(64.913 \text{ kPa})(1.8 \text{ m})}{2(0.03 \text{ m})} = 1.947 \text{ MPa} \quad (3.24)$$

$$\text{FS}_\tau = \frac{17.05 \text{ MPa}}{1.947 \text{ MPa}} = 8.76 \quad (3.25)$$

CONCLUSION

The ellipsoidal shell design for the pressure bulkhead successfully meets the required stress limits with factors of safety exceeding 4 in all stress categories. The calculated stress values at the top and equator of the shell show that both tension and shear stresses are within allowable limits, ensuring structural integrity. The analysis of connection methods indicates that both riveting and adhesive bonding can provide adequate shear resistance, with factors of safety exceeding the minimum requirements. Between our two options, the adhesive bonding option provides a higher factor of safety.

| A

APPENDIX A

A.1 Rough Work

AERE 426 - Lab Assignment 9

$$① T = 15.04 - 0.00649(13100_m) = -69.979,$$

$$P_{ext} = (101.325 \text{ kPa}) \left(\frac{-69.979 + 273.1}{288.08} \right)^{5.25588} = 16.147 \text{ kPa},$$

$$P_{in} = 0.8(101.325 \text{ kPa}) = 81.06 \text{ kPa},$$

$$\Delta P = P_{in} - P_{ext} = 64.913 \text{ kPa},$$

$$8 \text{ plies: } [0^\circ, -45^\circ, +45^\circ, +90^\circ]_s \Rightarrow t = 8 \cdot (125 e^{-6} \text{ m})$$

$$t = 0.001 \text{ m},$$

FS
↑

$$* \text{Top: } \sigma_\phi = \sigma_\theta = \frac{4(64.913 \text{ kPa})(1.8 \text{ m})^2}{26(0.001 \text{ m})} = 290 \times 10^3 \text{ kPa} \Rightarrow b = 1.450 \text{ m}$$

$$\sigma_\phi = \sigma_\theta = (290 \text{ MPa})/4 = 72.5 \text{ MPa},$$

$$\sigma_z = -81.06 \text{ kPa} = -0.0811 \text{ MPa},$$

$$* \text{Equator: } \sigma_\phi = \frac{(64.913 \text{ kPa})(1.8 \text{ m})}{2(0.001 \text{ m})} = 58.422 \text{ MPa},$$

$$\sigma_\theta = \frac{(64.913 \text{ kPa})(1.8 \text{ m}) \left(1 - \frac{(1.8 \text{ m})^2}{2(1.450 \text{ m})^2} \right)}{(0.001 \text{ m})} = 26.814 \text{ MPa},$$

$$\sigma_z = -0.0811 \text{ MPa},$$

$$* \text{Top: } \tau_{max} = \frac{72.5 \text{ MPa} - (-0.0811 \text{ MPa})}{2} = 36.291 \text{ MPa},$$

$$* \text{Equator: } \tau_{max} = \frac{58.422 \text{ MPa} - (-0.0811 \text{ MPa})}{2} = 29.252 \text{ MPa},$$

$$FS_o = \frac{290 \text{ MPa}}{26.814 \text{ MPa}} = 10.8,,$$

* Equator:

$$(2) \text{ Top: } FS_{\phi} = \frac{290 \text{ MPa}}{72.5 \text{ MPa}} = 4.00,,$$

$$FS_{\phi} = \frac{290 \text{ MPa}}{58.422 \text{ MPa}} = 4.96,,$$

$$FS_c = \frac{240 \text{ MPa}}{0.0811 \text{ MPa}} = 2960,,$$

$$FS_c = 2960,,$$

$$FS_r = \frac{160 \text{ MPa}}{36.291 \text{ MPa}} = 4.41,,$$

$$FS_r = \frac{160 \text{ MPa}}{29.252 \text{ MPa}} = 5.47,,$$

$$(3) m^2 \text{ rivets} = \frac{2\pi(1.8 \text{ m})}{(0.05 \text{ m})} = 226.195 \Rightarrow 227 \text{ rivets},,$$

$$F_{\text{rivet shear}} = \frac{\pi(1.8 \text{ m})^2(64.913 \text{ kPa})}{227 \text{ rivets}} = 2.911 \text{ kN} = 654.419 \text{ lbs},$$

$$F_{n.s., FS} = 4(654.419 \text{ lbs}) = 2617.68 \text{ lbs}$$

Chosen rivet: Ti-6Al-4V & Alloy Steel with $D = 3/16 \text{ in}$
and $F_u = 262.3 \text{ lbs}$

$$FS_r = \frac{262.3 \text{ lbs}}{654.419 \text{ lbs}} = 4.01,,$$

$$(4) \tau_{\text{adhesive}} = \frac{\pi d^2 \cdot \Delta P}{2\pi d \cdot W_{ad}} = \frac{(64.913 \text{ kPa})(1.8 \text{ m})}{2(0.03 \text{ m})} = 1.947 \text{ MPa},,$$

$$FS_r = \frac{17.05 \text{ MPa}}{1.947 \text{ MPa}} = 8.76,,$$